WEEK 4 • DAY 1

Machine Learning Foundations

Learn • Practise • Explain • Debug

Date	26-Oct-2026	**Duration**	5 hours
Difficulty	Intermediate	**Deliverable**	ML problem classification sheet

1. Today's Learning Goals

- Explain Machine Learning (ML) and how it differs from traditional rule-based programming.
- Distinguish supervised learning from unsupervised learning.
- Identify features (inputs) and labels (targets) in a dataset.
- Explain why data is split into training and test sets.
- Classify business problems by the ML approach they may need.
- Spot common problem-framing mistakes and improve the explanation.

2. Learn the Concepts in Simple Words

Machine Learning: A way to build systems that learn patterns from examples and use those patterns to predict outcomes or discover groups. Example: learn from past transactions to flag suspicious new transactions.

Traditional programming vs ML: Traditional programming often applies rules written by a developer: input + rules → output. In ML, an algorithm learns patterns from examples and uses them on new data.

Supervised learning: Training examples include a known target answer (label). Common tasks are classification (predict a category) and regression (predict a number).

Unsupervised learning: There is no supplied target label. The algorithm looks for structure or patterns, such as groups of customers with similar behaviour.

Features: Input columns used by a model. For a house-price model, features might include area, location, and number of bedrooms.

Label / target: The answer the model is trained to predict in supervised learning. For the house example, the label is the known sale price.

Training set: The portion of data used to train or fit the model.

Test set: A separate portion kept aside to estimate how the model performs on data it did not train on. Avoid repeatedly tuning decisions against the test set.

Important reminder: Not every problem needs ML. A clear rule, database query, or dashboard may be simpler, cheaper, and easier to explain.

3. Suggested 5-Hour Learning Plan

Time	Activity
Hour 1	Learn ML basics and compare rule-based programming with ML.
Hour 2	Study supervised vs unsupervised learning, classification vs regression.
Hour 3	Practise identifying features, labels, training data, and test data.
Hour 4	Complete the 10 business-problem classification exercise.
Hour 5	Debug common mistakes, explain answers in your own words, and review submission.

4. Quick Practice: Features and Labels

Example: predicting whether a customer will renew a subscription.

Possible features: months as a customer, number of logins, support tickets, plan type.

Label: renewed (Yes/No), if historical renewal outcomes are available.

For each scenario below, identify possible features and the label (if one exists). If no label is supplied, explain what patterns or groups you might look for.

5. Main Hands-On Activity: Classify 10 Business Problems

For each scenario, choose a starting approach: supervised classification, supervised regression, unsupervised clustering, or not necessarily ML / rule-based analysis. Explain your choice in 1–3 sentences. Real choices depend on the business goal and available data.

1. Fraud flag — A payments company has historical transactions marked fraudulent or legitimate and wants to flag new transactions for review.

2. Sales forecast — A retailer wants to estimate next month's sales amount using historical sales and relevant business data.

3. Customer groups — A business has customer purchase and engagement data but no predefined categories. It wants to discover groups with similar behaviour.

4. Subscription renewal — An app has historical records showing whether each customer renewed and wants to predict renewal for current customers.

5. House price — A property platform has past sale records and wants to estimate a property's likely sale price.

6. Duplicate invoice check — A company wants to flag an invoice if its invoice number already exists for the same vendor in its system.

7. Product recommendations — An online store wants to discover products often viewed or purchased together, without a manually supplied target label.

8. Support ticket priority — A support team has past tickets labelled urgent or non-urgent and wants to predict the priority of new tickets.

9. Inventory demand — A shop wants to predict the number of units of a product likely to be sold next week.

10. Spending patterns — A financial app wants to group users by spending behaviour but has no existing segment labels.

Classification Worksheet

#	Chosen approach	Possible features / data	Label (if any)	Reason / assumptions
1	Supervised Classification	Amount, location, transaction time	Fraud / Legitimate	Past transactions have labels. Predicts fraud.
2	Supervised Regression	Past sales, discounts, season	Next month's sales amount	Predicts a number using past data.
3	Unsupervised Clustering	Purchase history, spending, visits	No label	Groups similar customers.
4	Supervised Classification	Logins, plan type, usage	Renewed: Yes / No	Predicts whether a customer will renew.
5	Supervised Regression	Area, location, bedrooms	House price	Predicts a numeric price.
6	Rule-Based Analysis	Vendor ID, invoice number	No ML label	Checks if the same invoice already exists.
7	Unsupervised Learning (Association Rules)	Products viewed, products purchased	No label	Finds products often bought together.
8	Supervised Classification	Ticket text, issue type, keywords	Urgent / Non-urgent	Uses labelled tickets to predict priority.
9	Supervised Regression	Past sales, season, stock history	Units sold next week	Predicts the number of units needed.
10	Unsupervised Clustering	Spending amount, categories, frequency	No label	Groups users with similar spending habits.

Important for Question 7: Association rules are a type of unsupervised learning used to find products bought together. This is more accurate than clustering for this particular question.

6. Training vs Test Data: Mini Exercise

Imagine you have 1,000 historical customer records with known renewal outcomes.

1. Suggest a reasonable train/test split and explain your choice. An 80/20 split is a common starting point, not a universal rule.

Answer: Use an 80/20 split: 800 records for training and 200 for testing. It provides enough data to train and test the model.

2. Explain what the training set is used for and what the test set is used for.

Answer: The training set teaches the model patterns. The test set checks how well the model performs on unseen data.

3. Name one way information could accidentally leak from the test set into training or model decisions.

Answer: Data leakage happens when test data is used during training or to tune the model.

4. If records are time-dependent, explain why a time-based split may be more appropriate than a random split.

Answer: It trains the model on older records and tests it on newer records, which better represents real-world predictions.

7. Debugging: Identify and Improve the Mistakes

Mistake A: A student says: “We want to predict next month's sales amount, so this is classification.”

Your task: Correct the problem type and explain why.

Correction: This is regression because we predict next month's sales amount, which is a number.

Mistake B: A dataset has age, monthly spend, and city. A student says age is the label simply because it is the first column.

Your task: Explain how a target is chosen and what information is missing.

Correction: The target depends on what we want to predict. The first column is not automatically the target. We need to know the business goal.

Mistake C: A student trains a model on all records and reports its score on those same records as proof it works well on new customers.

Your task: Explain why this is misleading and how a test set helps.

Correction: The score can be misleading because the model has already seen those records. Use a separate test set to check performance on new data.

Mistake D: A team uses clustering but claims it can directly predict customer default even though no default labels were supplied.

Your task: Explain the difference between discovering groups and predicting a known target.

Correction: Clustering groups similar customers without labels. Predicting customer default requires known default outcomes to train a supervised model.

8. Explain in Your Own Words

1. What is ML? Explain it to a friend who has never coded.

Answer: Machine Learning (ML) is a branch of Artificial Intelligence that allows computers to learn patterns from data and make predictions without being programmed for every situation.

2. What is the difference between supervised and unsupervised learning?

Answer: Supervised learning uses labeled data with correct answers, while unsupervised learning uses unlabeled data to find patterns or groups.

3. What is the difference between a feature and a label? Give an example.

Answer: Feature: An input detail used by a machine learning model to make a prediction.
Label: The target answer that the model learns to predict.
Example: To predict a student's result:
- Features: Study hours and attendance.
- Label: Pass or Fail

4. Why should we test a model on data it did not train on?

Answer: We test a model on data it did not train on to check how accurately it can make predictions on new, unseen

5. Give one business problem where simple rules are better than ML, and explain why.

Answer:A shop can give a 10% discount to customers who purchase items worth more than ₹5,000. Simple rules are better because they are easy to implement and do not require training data or an ML model.

9. Final Deliverable and Submission Rules

Submit ONE file: Week4_Day1_ML_Problem_Classification_YourName.docx or .xlsx

- Complete all 10 rows of the classification worksheet.
- Include possible features/data and label (if any) for each scenario.
- Answer the training/test mini exercise.
- Complete all four debugging questions.
- Write explanations in your own words and state any assumptions.
- Be ready to explain two of your classifications to the trainer.

10. Self-Check Before Submission

- I can explain ML vs rule-based programming.
- I can distinguish classification, regression, and clustering.
- I can identify possible features and labels.
- I understand why training and test data should be separated.
- I classified all 10 scenarios and explained my reasoning.
- I corrected the debugging mistakes and can explain my answers.

Assessment Rubric (100 Marks)

Assessment area	Marks
10 problems: approach + reasoning	40
Features, labels, and assumptions	20
Training/test mini exercise	10
Debugging and improvement	20
Own-words explanations and clarity	10
TOTAL	**100**

Trainer note: Accept well-reasoned alternatives when the student states sensible assumptions. Recommendation and fraud problems can be approached in different ways depending on labels, data, and business goals.